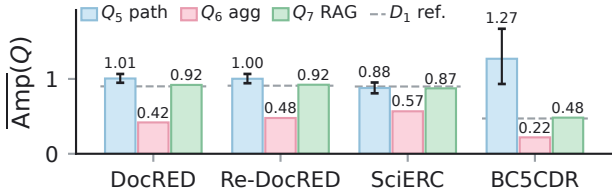


(a) Query-topology amplification



(b) Query-aware F1 gain at matched budgets

	local				multi-hop			
	30%	50%	70%	90%	30%	50%	70%	90%
DocRED	+0.15	+0.18	+0.15	+0.03	+0.07	+0.11	+0.10	+0.03
Re-DocRED	+0.16	+0.21	+0.16	+0.03	+0.20	+0.24	+0.16	+0.06
SciERC	+0.01	+0.03	+0.06	+0.04	+0.11	+0.12	+0.12	+0.05
BC5CDR	+0.17	+0.02	0	0	+0.16	+0.05	0	0

review budget